

# Bibek Koirala

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## SUMMARY

Computer Science MS (4.0 GPA) and graduate research assistant specializing in computer vision, deep learning, and applied machine learning. Experience spans instance segmentation, retrieval-augmented generation (RAG) with LLMs, and full-stack ML pipelines. Background combines academic research, six years of disciplined software engineering and R&D leadership in the Nepali Army, and prior work in healthcare data analytics. First-author paper accepted for oral presentation at IEEE ICMLA 2025.

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++, Java, SQL

**Machine Learning & Deep Learning:** PyTorch, TensorFlow, scikit-learn, Transformers, CNNs, Vision Transformers, Mask R-CNN, ResNet, Object Detection, Instance Segmentation, Regression, Support Vector Machines, Decision Trees, Principal Component Analysis (PCA), Kernel Methods, Clustering

**NLP & LLMs:** Large Language Models (LLMs), Vision-Language Models (VLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, HuggingFace, FAISS, Sentence Transformers, Vector Databases, Agentic AI

**Data & Computer Vision:** OpenCV, NumPy, SciPy, Pandas, Matplotlib, Statistics, Convex Optimization

**Tools & Platforms:** Git, GitHub Actions (CI/CD), Docker, Linux, PostgreSQL, VS Code, Jupyter, LaTeX, Experiment Tracking, Model Versioning

## EDUCATION

**Southern Illinois University Carbondale — M.S., Computer Science**

Aug 2024 – Present

*Graduate Research Assistant, Computer Vision and Object Detection*

- GPA: 4.0 / 4.0
- **Coursework:** Machine Learning and Soft Computing, Artificial Intelligence, Deep Learning, Data Mining, Natural Language Processing, Parallel and Distributed Computing, Advanced Database Systems, Cryptography and Network Security

**Pulchowk Campus, IOE, Tribhuvan University — B.E., Computer Engineering**

Nov 2012 – Sep 2016

- Awarded full merit scholarship; ranked in the top 1% of approximately 15,000 entrance exam applicants.
- **Coursework:** Data Structures and Algorithms, Operating Systems, Object-Oriented Programming, Computer Organization and Architecture, Signal Processing, Big Data, Database Systems, Computer Security, Software Development, Artificial Intelligence

## PROFESSIONAL EXPERIENCE

**Graduate Research Assistant — Southern Illinois University Carbondale**

Aug 2024 – Present

*Department of Computer Science, Carbondale, IL*

- Developed an end-to-end Deep Learning pipeline for instance segmentation on time-stamped microscopy images of bacterial cultures under varying backgrounds, achieving mIoU of 0.91–0.98 across growth stages.
- Improved early-stage detection from 3-hour to 2-hour incubation, reaching mAP@0.5 of approximately 0.95; first-author paper accepted for **oral presentation at IEEE ICMLA 2025**.
- Built end-to-end research codebases in PyTorch with reproducible training, evaluation, and experiment-tracking workflows.

**Army Officer — Software Development and R&D, Nepali Army**

May 2018 – Dec 2023

*Kathmandu, Nepal*

- Engineered automated artillery fire-control software, reducing targeting error from 50 m to 10 m at 1000 m range and cutting computation latency from minutes to microseconds through optimized numerical pipelines.
- Designed and developed a Windows desktop dashboard with a relational database backend for managing troop leave and training records, replacing paper workflows and improving administrative throughput.
- Developed decision-support software to compute logistical requirements for Bailey bridge deployment, improving accuracy and field deployment speed under operational constraints.

- Completed multiple leadership and technical training programs emphasizing systems thinking, operational reliability, and disciplined execution under pressure.

#### **Data Analyst — Morang Hospital Pvt. Ltd.**

Dec 2016 – Feb 2018

*Pathari, Nepal*

- Maintained hospital inventory and patient records databases using PostgreSQL; authored queries to track pharmaceutical procurement and sales transactions.
- Performed exploratory data analysis on patient admissions, billing, and inventory datasets to surface operational trends and support data-driven decisions.

### **SELECTED RESEARCH PROJECTS**

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#### **Bacteria Detection** — [github.com/vbek/bacteria-detection](https://github.com/vbek/bacteria-detection)

- Developed an end-to-end instance segmentation framework for rapid bacterial detection in microscopy images.
- Achieved significant improvement in early detection capability, reducing effective incubation requirement from 3 hours to 2 hours while maintaining high mAP@0.5 performance.
- IEEE Xplore: [ieeexplore.ieee.org/document/11471447](https://ieeexplore.ieee.org/document/11471447)

#### **SEEKBOT: Full-Stack RAG Chatbot** — [github.com/vbek/SEEKBOT](https://github.com/vbek/SEEKBOT)

- Built a 10-node LangGraph pipeline with a local FAISS vector store and HuggingFace Inference API (Llama 3.1 8B and Qwen 2.5 72B) for semantic search across PDF, DOCX, CSV, Markdown, and web content; persistent sessions and fully local embeddings via sentence-transformers.

#### **Plant Disease Detection (Comparative Deep Learning Study)** — [github.com/vbek/Plant-Disease-Detection](https://github.com/vbek/Plant-Disease-Detection)

- Compared CNN (34.7M params), fine-tuned ResNet50 (9.9M trainable), vanilla ResNet (36.4M), and Vision Transformer (57.5M) on PlantVillage; accuracies of 99.0%, 99.6%, 98.9%, and 97.9% respectively.

#### **News Credibility Detection** — [github.com/vbek/News-Credibility](https://github.com/vbek/News-Credibility)

- Built an NLP and ML framework classifying news articles using linguistic, semantic, and source-based features; achieved 97% precision and 98% recall on benchmark datasets.

#### **Remittance and Inflation Dynamics** — [github.com/vbek/Inflation-Dynamics](https://github.com/vbek/Inflation-Dynamics)

- Econometric and ML analysis of countries with remittance above 5% of GDP (1960–2023), using clustering to examine relationships among macroeconomic indicators.

#### **Crime Victimization Pattern Analysis** — [github.com/vbek/Victim-Crime-Perception](https://github.com/vbek/Victim-Crime-Perception)

- Analyzed a sparse, high-dimensional criminology survey to identify victimization patterns and cluster victims by reporting behavior, with results aligning with prior research.

#### **Military Leave Management Dashboard** — [github.com/vbek/Military-Leave-Management-Dashboard](https://github.com/vbek/Military-Leave-Management-Dashboard)

- Windows-based GUI dashboard with database-backed information system for troop leave and training records, enabling a fully paperless administrative workflow.

### **LEADERSHIP AND ACTIVITIES**

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- Captain, Pulchowk Campus Cricket Team, 2015 and 2016; competed at the district level in 2016 and 2017.
- Active participant in volleyball, chess, badminton, and running events.
- Six years of military service developed leadership, team coordination, and operational discipline.